

BOW VALLEY, AB, Canada

WMO: 712320

Lat: 51.083N

Lon: 115.066W

Elev: 1298

StdP: 86.68

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-30.7	-26.2	-34.2	0.2	-30.5	-29.7	0.3	-25.8	11.1	-1.6	10.2	-1.1	0.5	50	0.659

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.4	28.8	14.5	26.8	14.0	24.7	13.3	16.6	24.3	15.4	23.3	14.4	22.3	3.4	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.9	11.6	19.1	12.5	10.6	17.5	11.4	9.8	16.3	51.2	24.4	47.6	23.4	44.4	22.3	21.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.9	7.8	6.9	-36.8	32.1	3.8	2.0	-39.5	33.6	-41.7	34.7	-43.9	35.8	-46.6	37.3		
(5)			-36.9	18.6	3.7	1.4	-39.6	19.6	-41.8	20.4	-43.9	21.2	-46.6	22.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.9	-6.2	-5.5	-1.8	3.2	8.0	11.9	15.5	14.6	10.3	4.6	-1.8	-6.5
	DBStd	10.00	9.11	8.31	7.89	5.03	4.43	3.33	3.18	3.42	4.59	5.15	7.34	8.34
	HDD10.0	2727	503	434	367	209	92	17	2	6	52	176	356	512
	HDD18.3	5286	761	668	625	455	322	195	97	123	241	424	604	771
	CDD10.0	501	0	0	1	4	29	74	174	147	62	10	1	0
	CDD18.3	18	0	0	0	0	0	2	9	7	1	0	0	0
	CDH23.3	729	0	0	0	0	14	55	324	267	68	0	0	0
	CDH26.7	184	0	0	0	0	1	12	94	67	11	0	0	0
(14) Wind	WSAvg	2.9	3.8	3.3	2.9	2.6	2.5	2.4	2.4	2.2	2.5	3.0	3.4	3.9
(15) Precipitation	PrecAvg	564	19	18	32	42	78	109	59	61	61	37	30	18
	PrecMax	791	46	34	77	125	144	357	187	153	147	76	85	52
	PrecMin	369	0	0	6	7	16	27	16	0	6	13	9	5
	PrecStd	116	11	10	16	28	33	71	38	34	37	15	17	10
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.0	11.1	15.3	20.4	25.2	28.5	32.0	31.0	28.4	20.2	14.3	10.2
		MCWB	4.9	3.7	5.5	8.7	11.4	14.2	15.5	14.6	13.4	10.1	6.4	4.0
	2%	DB	8.2	8.7	11.9	17.1	22.5	24.6	29.2	28.6	25.3	16.9	11.4	7.4
		MCWB	2.5	2.4	4.2	7.3	10.3	13.0	15.4	14.0	12.4	8.4	5.2	2.1
	5%	DB	6.3	6.7	9.7	14.3	19.9	22.2	27.0	26.5	22.7	14.7	9.2	5.4
		MCWB	1.3	1.1	3.1	5.8	9.5	12.1	14.6	13.9	11.6	7.3	3.9	0.8
	10%	DB	4.5	4.5	7.7	11.8	17.3	20.0	24.6	24.1	19.5	12.5	7.0	3.4
		MCWB	0.2	-0.3	2.0	4.5	8.5	10.9	14.0	13.4	10.4	6.0	2.3	-0.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.3	4.3	6.2	9.3	12.9	16.4	19.1	17.4	14.9	10.9	7.1	4.5
		MCDB	10.0	10.1	13.3	18.8	21.7	23.4	26.2	24.7	24.6	18.8	12.6	9.6
	2%	WB	3.0	2.7	4.8	7.8	11.4	14.5	17.1	16.1	13.3	9.2	5.6	2.4
		MCDB	7.5	8.1	11.0	15.9	19.5	21.9	24.8	23.7	22.7	15.8	10.8	6.8
	5%	WB	1.5	1.2	3.5	6.3	10.2	13.0	15.9	15.0	12.1	7.7	4.1	0.9
		MCDB	5.7	6.1	9.1	13.4	18.1	20.3	23.8	23.5	20.9	13.8	9.0	5.1
	10%	WB	0.3	-0.1	2.2	4.8	9.0	11.8	14.9	14.0	10.8	6.4	2.5	-0.4
		MCDB	4.4	4.1	7.5	11.0	16.7	18.1	22.4	21.9	18.5	11.9	6.7	3.2
(35) Mean Daily Temperature Range	MDBR		9.7	11.3	11.6	11.9	13.0	12.7	15.4	15.7	13.8	10.7	8.6	8.6
	5% DB	MCDBR	8.5	10.9	12.3	16.4	18.4	18.1	20.8	20.9	20.1	14.1	10.0	8.1
		MCWBR	5.4	6.7	7.0	8.9	9.1	8.7	9.0	9.0	9.2	7.7	6.2	5.6
	5% WB	MCDBR	8.3	11.0	12.2	15.4	16.5	16.1	17.9	18.5	18.8	13.7	10.6	8.4
		MCWBR	5.6	7.1	7.3	8.6	8.6	8.3	8.9	9.3	9.2	7.8	6.9	6.1
(40) Clear-Sky Solar Irradiance	taub		0.251	0.265	0.286	0.337	0.346	0.359	0.348	0.362	0.320	0.282	0.257	0.240
	taud		2.437	2.427	2.418	2.291	2.362	2.366	2.409	2.383	2.482	2.574	2.516	2.426
	Ebn at Noon		807	886	924	903	908	896	900	868	875	859	796	764
	Edh at Noon		56	75	92	118	115	116	110	107	86	65	52	48
(44) All-Sky Solar Radiation	RadAvg		1.12	2.07	3.27	4.55	5.29	5.59	6.28	5.21	3.67	2.28	1.21	0.88
	RadStd		0.07	0.14	0.22	0.32	0.44	0.46	0.45	0.37	0.42	0.22	0.08	0.05

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (48 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page